Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
Rectified gain @ k=96 (kcal/mol)
Predicted most stabilizing per domain
8.67
13.25
22.47
4.45
9.10
14.23
5.03
9.45
15.07
3.64
6.94
10.54
3.81
8.46
12.90
4.20
8.34
13.76
4.25
4.88
6.01
4.23
5.46
6.91
4.09
5.77
6.42
1.94
5.05
10.51
1.57
4.98
11.93
21.58
30.24
44.18
19.29
33.42
43.28
20.02
32.35
43.63
19.93
32.54
43.54
17.37
33.71
44.92
18.60
31.55
45.86
38.64
31.21
26.15
38.42
30.06
27.52
34.94
32.06
29.00
12.91
24.48
58.61
9.24
27.24
59.52
Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
Δ volume ≤ −1
−1 < Δ volume ≤ 1
Δ volume > 1
Top 96 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,421 single mutants); gain = sum of max(ddG, 0) over every pick, whatever its predicted sign; boxes: interquartile range with darker colored median line; dots: domains; dark bars & annotations: mean over domains; dashed lines: oracle mean.